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Anonymization for Patient data

@ Neural Wave Hackathon

This report describes an iterative, evaluation-driven development process. Several detection strategies were built, measured against a common evaluation framework and progressively consolidated into a single offline anonymizer.

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Problem statement

Recorded doctor–patient dialogues and clinical notes must be stripped of identifying information before any use with cloud-based language models, while preserving the information required to assign the original ICD-10 codes. Inference must run on-premises within 4 GB of VRAM, and the codes themselves must remain unchanged. The central difficulty is ambiguity: *Parkinson* may denote a disease or a surname, and *golfing* may be a personal detail or the documented cause of an injury. Privacy protection, preservation of medical information, and originality and scalability are weighted equally.

Methodology: evaluation before integration

Before committing to any technique, the team established a shared evaluation framework that tracks the challenge's metrics: silver-standard labels, fixed data splits, a single scorer and an acceptance gate. Design choices were tuned on one split and progress measured on another, while a held-out test set of unseen formats was scored only once. Every technique discussed in this report, whether regular expressions, named-entity recognition, language models, fine-tuning or learned classifiers, was retained or discarded on the basis of measured results rather than expected benefit.

Development process at a glance



1 Zero-shot baseline with a small language model

The first experiment assessed whether a small model could perform the task end to end. **Qwen3.5-2B** was run zero-shot through Ollama and vLLM (temperature 0) on segments of about 2,000 characters. It returned personal information as structured JSON spans, which were then replaced deterministically with category tags. Additional components generalized ages and dates (for example "forty-something" or "mid 2020s"), retained terms occurring in the encounter's own ICD-10 descriptions, and logged each response so that policy changes could be re-evaluated without further model calls. With FP8 weights, peak memory remained within 4,000 MiB.

Findings. The model handled dates well (91% removed), but because it processes each segment independently it showed systematic weaknesses:

- names were not linked across turns and fields, so only 85.7% of patient names were removed;
- clinical terms were misclassified as organizations or locations ("diabetes" was tagged as an organization; 326 organization tags against 37 actual organizations);
- 188 of 292 predicted dates were in fact doses or durations, and 8–13 of 400 responses produced malformed JSON.

Baseline: 75.2% of privacy-related information removed; medical concept recall 94.3%.

2 Rule-based detection extended with NER

The second phase inverted the approach on the premise that **regular identifiers are best handled by deterministic rules**. These comprised context-aware regular expressions (dates of birth in more than six formats, record numbers, spoken ages, titles followed by names) and dictionary lookups against public reference lists (ICD-10-CM terms, drug names, census name lists, GeoNames place names).

Named-entity recognition was then layered on top of the rules:

- **GLiNER2**, with additional "medication" and "medical concept" labels, so that drugs and diseases no longer compete with organization labels;
- **deid-RoBERTa** and **GLiNER-PII** as alternative recognizers.

Holdout, 30 encounters	Direct identifiers removed	Quasi-identifiers removed	Clinical spans kept	Runtime / enc.
Rules only	97.1%	67.2%	100%	0.04 s
Rules + GLiNER2	98.5%	75.0%	100%	~1.9 s

Rules alone accounted for most of the removal at a fraction of the cost. Learned models added value chiefly on quasi-identifiers, which rules cannot capture.

From Parallel Prototypes to a Unified Pipeline

3 Parallel prototypes and comparative assessment

Four prototypes tackled the same problem: an **LLM-first** harness (extraction, then generalization), a **shield-first** rule-based harness (clinical content protected before identifiers are removed), a **GLiNER2** pipeline with cross-field entity resolution, and a fully AI-generated implementation. Merged labels and a **single scorer** (reproducing both originals to four decimals) made them comparable:

Approach (validation, 67 encounters)	Privacy-related info removed	Clinical traps kept
Unified pipeline (final)	84.6%	99.1%
LLM-first harness	77.1%	79.6%
Shield-first harness with NER	81.5%	99.4%

Rules plus an identity graph won on names (100% of name tokens removed); the language model won on occupations and life events. The two were complementary.

4 Mid-term evaluation

A **simple harness**, scored by the organizers on their non-public dataset:

0.734	0.0018	0.997
i.a proportion of removed privacy-related information	ii.a F1-Score ICD-10 code prediction, Δ (orig. - anon.)	ii.b medical concept recall

Medical information was almost fully preserved, so **privacy had the most headroom**: occupations (17 of 33 remained), organizations, low-risk details, unseen formats.

Organizers' metrics on their data; all other figures are internal and not directly comparable.

5 Unified architecture: eight stages, one audit trail

01 Input
Strict JSONL, one encounter per line, validated against the output contract.
Why: fails closed; no record is skipped silently.

02 Normalize
One-to-one character view, same length, speaker tags masked.
Why: rules, NER and LLM spans merge without offset drift.

03 Shield
Locks clinical values, drugs, medical eponyms, terms supporting the ICD-10 codes and injury mechanisms.
Why: utility protected *by construction*.

04 Detect
Rules first (~50 ms); the local LLM sees only privacy-dense segments (work, home, family, travel).
Why: detectors ordered by cost; LLM where rules are blind.

05 Link
Per-encounter identity graph across turns and fields, incl. typos, possessives, titles.
Why: closes segment-level leaks.

06 Policy
Replace, generalize (ages to decade bands, dates to [DATE]) or keep, per category and preset.
Why: the trade-off is an explicit configuration.

07 Verify
Leak scan for every removed string, two repair rounds, contract re-checked.
Why: independent second pass; repairs only remove.

08 Publish
Atomic dated outputs plus a decision log (category, action, detector, reason).
Why: explains each decision (iv.b), stores no source text.

Shield-first backbone (shield, rules, identity graph, leak repair) plus the LLM-first strengths (targeted extraction, age generalization, response cache). Two modes: **rules** (CPU only) and **llm** (default: rules + qwen3.5:2b via Ollama, preset *privacy*). A 147 MB container runs fully offline.

Gated optimization

A gate on the tune split ensures privacy never regresses: high-risk removal may not drop, quasi or low-risk removal must rise, traps and concept recall stay **≥ 99.5%**, ICD-10 agreement stays within noise, codes byte-identical.

Feature (tune split)	Measured effect	Default preset
F0: code-evidence guard	low-risk removal 0.23 → 0.37; traps 100%	enabled
F2: targeted LLM detector	occupations still present 17 → 11 of 33	enabled
privacy preset	i.a 90.1% removed (rules), traps 100%	default
F4: two-signal agreement	low-risk removal 0.52 → 0.38	opt-in only

Fine-tuning a decision policy (SFT + PPO)

Qwen3.5-2B was trained to **decide what to remove, not to rewrite the text**. It returns its decisions as JSON and ordinary code applies them, so the ICD-10 codes are never altered. Training used about 1,000 labelled examples; PPO rewarded each correct decision and penalized removing too much. The model passed the gate, but its gains were within measurement noise, so it was **not used**.

6 Learned shield

A "System 1" classifier to protect medical text the rules don't know. More memory, **no measured gain**: excluded.

7 Unseen formats

ICU-note postcodes and dates clashed with numeric shields; fixed, then scored once. Traps kept 95.3%, below the 99.5% floor: **the main open gap**.

Internal evaluation framework

Reference labels. 231 files, three sources, 26 disagreements resolved; never seen by the pipeline.

Splits. Tune (140) for design decisions, validation (67 ACI-Bench encounters), test (24 unseen hospital documents, scored once).

Main metrics (i.a, ii.b). Proportion of removed privacy-related information, by risk tier; medical concept recall; clinical traps kept.

F1-Score ICD-10 code prediction (ii.a). LLM judge (majority of 3 Claude Sonnet 5.5 agents) and a decision tree, original vs anonymized.

Adversarial privacy attacks (iv.a). Six attacks, from NER rescan to an LLM red-team agent; proportion of recovered information.

Robustness (iv.e). Same metrics as the main task on 192 variants, 8 held-out hospital formats, 8 Kaggle suites.

Evaluation Results: ACI-Bench and Kaggle Datasets

Datasets and metrics

ACI-Bench: the evaluation split given by the hackathon challenge, 67 doctor–patient dialogues and notes with ICD-10 codes. **Kaggle**: open-source medical and synthetic data the team collected (discharge summaries, scribe transcripts in five languages, long dialogues, NBME notes, PMC case reports, essays), without ICD-10 codes and with partial privacy labels.

i.a Privacy-related information removed. Input: original and anonymized text, 538 labelled private items. Share of their occurrences with no distinctive part left. Direct identifiers: names, dates, IDs, contacts.

ii.b Medical concept recall. Input: original and anonymized text, 6,607 labelled medical terms. Share of their occurrences still present.

ii.a F1-Score ICD-10 code prediction. Codes predicted from the original and the anonymized note, scored against the ACI-Bench gold codes: F1 = 2TP / (2TP + FP + FN); Δ = anonymized − original.

ii.a coders. LLM as a judge: Claude Sonnet 5.5, three passes, majority vote per note. **Decision tree**: trained on the challenge's ACI-Bench training encounters and their codes; predicts 3-character categories.

Same codes. Codes from anonymized text scored against codes from the original; 1 = identical. Used for Kaggle (no gold codes).

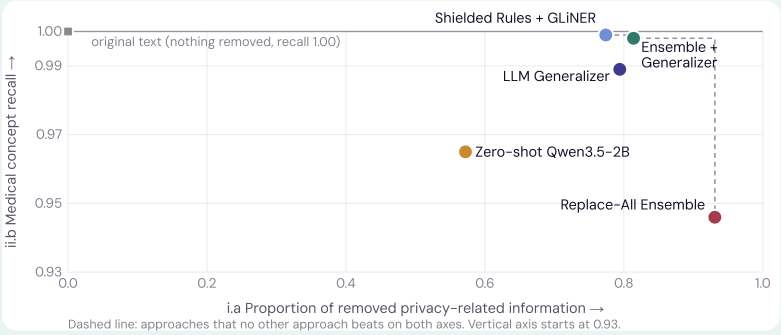
iv.a.ii Recovered information. A blinded LLM agent reads the anonymized text and guesses name, age, place and job; share of correct guesses. **iv.c**: first 10 encounters; VRAM incl. Ollama.

Main results on the ACI-Bench evaluation split (67 encounters)

Approach	i.a Privacy-related information removed	i.a Direct identifiers removed	ii.b Medical concept recall	ii.a F1-Score ICD-10 code (LLM judge)	Δ vs. original (LLM judge)	ii.a F1-Score ICD-10 category (tree)	Δ vs. original (tree)	iv.a.ii Proportion of recovered information
Ensemble + Generalizer	0.814	0.926	0.998	0.554	+0.023	0.689	+0.000	0.125
Replace-All Ensemble	0.931	0.982	0.946	0.536	+0.005	0.689	+0.000	0.092
LLM Generalizer	0.794	0.944	0.989	0.551	+0.020	0.689	+0.000	0.178
Shielded Rules + GLiNER	0.774	0.926	0.999	0.573	+0.042	0.689	+0.000	0.138
Zero-shot Qwen3.5-2B	0.572	0.705	0.965	0.540	+0.008	0.689	+0.000	0.441

Ensemble + Generalizer = the final pipeline, combining the next two (LLM mode, privacy preset); **Replace-All Ensemble** = regular expressions, name lists, deid-RoBERTa and GLiNER, replacing every detected span; **LLM Generalizer** = turns private details into general ones (exact age → age range); **Shielded Rules + GLiNER** = regular expressions and rule-based shields with GLiNER; **Zero-shot Qwen3.5-2B** = the model alone, prompted to output the anonymized text. Original text: judge F1 0.531, tree F1 0.689, recovered 0.822.

Trade-off between privacy (i.a) and medical concept recall (ii.b)



What the trade-off shows

Each point is one approach: further right means more privacy-related information removed, higher means more medical concepts kept. **Ensemble + Generalizer** sits at the top right: it keeps almost every medical concept (0.998) while removing 0.814 of the private information.

Replace-All Ensemble removes the most (0.931), especially occupations and organization names, but replacing every detected span also removes medical terms, so its concept recall drops to 0.946.

Efficiency optimization (bonus iv.c)

Approach	iii Peak VRAM	Within 4 GB	i Runtime per encounter	Load time
Ensemble + Generalizer	2,850 MB	yes	4,131 ms	9.0 s
Replace-All Ensemble	3,334 MB	yes	1,801 ms	34.3 s
Shielded Rules + GLiNER	1,966 MB	yes	629 ms	23.3 s
Zero-shot Qwen3.5-2B	2,960 MB	yes	1,882 ms	7.9 s

NVIDIA RTX PRO 6000 Blackwell. **Without ICD-10 codes at inference (bonus iv.d):** Ensemble + Generalizer removes more privacy-related information (0.829 vs 0.814) with the same F1-Score ICD-10 code prediction (0.554).

Robustness on external Kaggle datasets (bonus iv.e)

Approach (i.a removed)	Disch.	Scribe EN	Scribe non-EN	Long dial.	NBME	PMC	Essays names	Essays IDs
Ensemble + Generalizer	0.89	1.00	0.98	1.00	0.81	0.96	0.84	0.78
Replace-All Ensemble	0.84	1.00	0.98	1.00	0.95	1.00	1.00	0.92
Shielded Rules + GLiNER	1.00	1.00	1.00	1.00	0.81	0.96	0.99	0.94
records per dataset	50	42	60	4	100	100	100	100

Labels are partial (NBME, PMC: age only), so scores show misses, not exact rates. Same codes (decision tree): **1.00 on every dataset for Ensemble + Generalizer**; Shielded Rules + GLiNER 0.98, Replace-All Ensemble 0.86 at worst.

Conclusions and limitations

- Protecting clinical content before detection proved more reliable than repairing it afterwards; small LLMs work best where rules are blind, not as the whole pipeline.
- We chose **Ensemble + Generalizer** as the final approach: it keeps far more medical concepts than Replace-All Ensemble (ii.b 0.998 vs 0.946) with a slightly higher F1-Score ICD-10 code prediction (ii.a 0.554 vs 0.536, within judge noise), at the cost of removing less privacy-related information (i.a 0.814 vs 0.931).
- Differences of about 0.10 in the proportion of recovered information (iv.a.ii) are within the variation of a single LLM adversary per set. Record linkage through the ICD-10 codes, which must be kept, makes k-anonymity unattainable.
- Limitations: all three ACI-Bench label sources were produced with LLM assistance; unseen formats score lower; some date forms are missed.